Grama Jyothi Rural Electricity Co-operative
Task 3

TASK 3 – Recommend and Defend the Final Design (K5)

Recommended Design Meter readings: uint16_t when the meter specification is exactly a 16-bit unsigned counter; validate input before conversion. Consumption: uint32_t, because rollover consumption may exceed the current reading and should be represented explicitly. Money: int64_t in paise for each bill and the aggregate total. Confirm that the maximum plausible district total fits the chosen type. Tariff constants: centralize them in one header or one tariff structure rather than repeating numeric values in multiple functions. Preprocessor: use include guards and #define for compile-time constants. Use a debug macro only for optional diagnostic output. Validation: check scanf results, numeric ranges, negative values, and unexpected reading decreases. Tariff revision with compiled constants: update the centralized tariff definitions, run all boundary tests, compile, and redistribute the tested executable. A configuration file can avoid recompilation, but it needs parsing, validation, error handling, and a clear policy for missing or invalid values. Configuration File Alternative and Complexity Reading t tariff entries at startup takes O(t) time and O(t) memory if all entries are stored. For a small tariff table, this overhead is usually small compared with processing hundreds of thousands of households. The benefit is that authorized tariff changes can be deployed without recompiling; the risk is malformed or inconsistent configuration, so validate slab order, non-negative rates, and units before accepting the file. Evidence to add: paste your actual Task 1 test results and Task 2 measurements here before final submission.